

Albert Rich 1949-2023

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1 Abstract

This first section comes mostly from an autobiographical account, published in 1998 . Albert D. Rich III was born in Altadena, California on April 26, 1949 to Albert D. Rich II and Alberta W. Rich. Being almost ten years younger than his three siblings, he was probably an unexpected surprise to his parents. At age 7, the family moved to Nacogdoches, Texas where Albert was raised. His father was a chemical engineer in the vegetable oil bleaching clay industry. Because of his mother’s love of travel, she started a successful travel agency. Albert was active in the Boy Scouts, but hated competitive sports.

Albert graduated from Nacogdoches High School in 1967, and won a Navy ROTC scholarship to the University of Texas. There he learned various programming languages, but immediately latched onto LISP as the language most in tune with his way of thinking.

After studying Bertrand Russell’s *Principia Mathematica*, Albert became interested in automating the process of proving theorems and generating interesting new ones. He collaborated with Laurent Siklossy, a professor of computer science, on a paper about a unification algorithm invented by Albert for proving theorems in the propositional calculus. In 1971, he received a BA in mathematics from UT.

Upon graduation, Albert was commissioned as an officer in the US Navy*. After a year of Navy nuclear power school, he served the remainder of his military commitment on the U.S.S. James Monroe, a ballistic-missile submarine home-ported in Honolulu, Hawaii. In June of 1976, he was honorably discharged in Newport-News, Virginia, but immediately went racing back to Hawaii.

Having some ideas for theorem proving that he wanted to implement, Albert bought a kit for an IMSAI 8080 computer and soldered it together[†]. This S-100 computer had 4K memory

*DJJ: It was a condition of his university scholarship that he serve in the US Navy after graduation.

[†]The computer was proudly kept by Albert, even after being replaced by modern laptops, and was donated early in 2024 to the University of Waterloo Computer Science museum

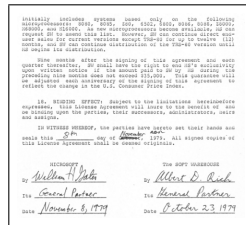


Figure 1: A fragment of the licence agreement between Microsoft and Soft warehouse

boards costing about \$500 each (several 1000 times more per byte than now). Having only enough money to populate one board with just 2K bytes of memory, Albert gained a real appreciation for a byte: a concept seeming unknown to the current generation of programmers.

The next step was to implement a LISP interpreter for the IMSAI. By hand-assembling machine-code and entering it from the front panel switches[‡], “first light” for the interpreter was achieved by the end of 1976 (at about the same time, Bill Gates and Paul Allen were writing their Basic interpreter for the Altair computer).

In 1977, David Stoutemyer, a professor at the University of Hawaii, heard that Albert had a LISP system running on an Intel 8080 based computer. Having done pioneering work in the field of computer algebra, David was interested in bringing these capabilities out of the M.I.T. and Stanford closets and making them available to the masses. At that time, LISP was the best language for implementing computer algebra systems, and so Albert and David joined forces to work on the first and only CAS that would run on CP/M based machines. In 1979, Albert and David formed Soft Warehouse, Inc. and began selling muLisp and muMATH.

A story from those early days concerns a visit to Hawaii by Bill Gates and Paul Allen. They were looking for software products that their newly formed company Microsoft could sell, in addition to their own software. While showing them muMath, Albert offered to take them swimming at Makapu’u beach. The waves must have been quite high that day, because the founders of Microsoft were knocked sideways and had trouble regaining their feet. Exactly how water-logged they were, and exactly how much assistance was needed from Albert for them to regain the shore is the stuff of legend, and varies with the telling. In spite of the experience, Microsoft added muMath to their catalogue, and were an agent for Soft Warehouse for a number of years. Albert preserved his copy of the contract.

Albert and Paul Allen became friends, and in 1983, when fighting Hodgkin’s lymphoma, Paul Allen chose to spend time with Albert and Joan in Hawaii. In later years, Paul Allen hosted several holiday cruises on chartered cruise liners, where Albert was invited to mix with celebrities such as Robin Williams and Sarah McLachlan.

Albert had a long-standing passion for diving, and when Soft Warehouse was purchased by Texas Instruments, Albert moved to the ‘Big Island’ from Honolulu. The Big Island attracted

[‡]DJJ: For the younger generation, this requires some explanation. In those primitive days, when you turned on your new computer, it did nothing but sit there showing a few lights. You had to load a program laboriously by setting a row of toggle switches on the machine console. To load one word of memory, you set a bank of switches to the required address in memory, and another bank to the memory contents. This word was then loaded, and you repeated this until your program was loaded.



Figure 2: Albert and his HumVee. Although the HumVee was mostly used for dive trips, this photo was actually taken 8000 feet up Mount Mauna Kea.



Figure 3: A montage showing Albert standing in front of typical lava-strewn land near Kona on the big island. The tree in the distance behind him, is shown in close-up next to his photo.

Albert because of its young geological age and low population density. This made the beaches and coral reefs interesting to explore, because of their pristine state. Albert bought a HumVee[§] so that he could reach remote dive sites. In addition to diving at sites around Hawaii, Albert made some dive trips to distant places, such as reefs between New Guinea and Australia.

The Hawaiian islands have a ‘wet side’ and a ‘dry side’ because of the mountain ranges that run down their centres. The tourist hotels are mostly built on the dry side, for example at Waikiki. In order to service the tourists, water is transferred from the wet side to the dry side, and used to create artificial “tropical paradises” with lush vegetation. Albert became interested in nature reserves in which the natural desert-like plants and animals could be preserved. Both on Oahu and the Big Island, he supported such reserves.

Keywords

Albert Rich, Biography

References

- [1] A. D. RICH, *Rubi (Rule-based Integrator)*,
<https://rulebasedintegration.org/about.html>

[§]The military-grade version, not the Hummer civilian spin-off.